

Run And Gun

If you've played games like *Halo 2*, *Command & Conquer: Generals*, or *Ghost Recon*, you've doubtless marvelled at the aircraft in the battle scenes. These planes are supposed to be flying at speeds of over Mach 10!

"Force Application and Launch from Continental United States" (a.k.a. the Falcon Program) is a research and joint development program of DARPA and the United States Air Force for a high-speed strike vehicle capable of attacking *anywhere on the planet* in less than two hours! Operating at Mach 11 and a cruise altitude of 90,000 ft, this mean machine is going to be a reusable two-stage system comprising an unmanned boost vehicle used to take off at high speeds, and a manned hypersonic aircraft used in the second stage for the strike.

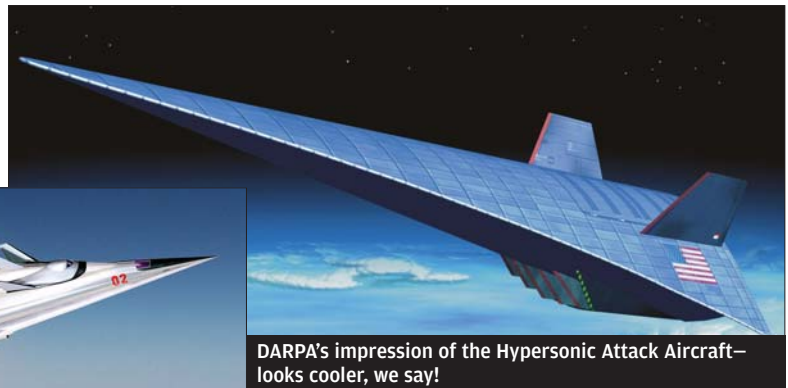
The unmanned boost vehicle has a gas-fuelled turbine engine. It requires a conventional runway and accelerates the strike vehicle, which is attached to the boost vehicle, to Mach 4 and 60,000 ft. Once this altitude and speed is reached, the manned strike vehicle then



An artist's impression of the Hypersonic Attack Aircraft

detaches itself from the primary boost vehicle and uses a scramjet-technology-based engine to reach its final cruise condition of Mach 11 at 90,000 ft. Schumacher just developed a complex... and with good reason!

The total range of the system is a purported 10,000 nautical miles. The hypersonic strike vehicle has an un-refuelled range of 5,000 nautical miles, which *very impressive*—Mumbai to Singapore in less than half an hour! The hypersonic attack aircraft has the capability of engaging a target very accurately, using precision-



DARPA's impression of the Hypersonic Attack Aircraft—looks cooler, we say!

guided missiles from a distance of 1,450 nautical miles away.

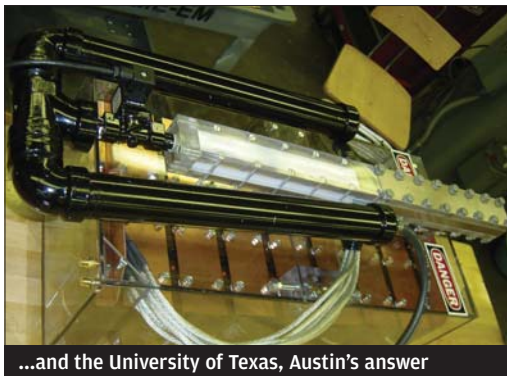
Expect to see it developed by end of 2007, and a demonstration by 2009.

Instant Gratification

This weapon was made famous in the *Quake* genre, which allowed players a one-shot-kill on opponents at the speed of light... now who wouldn't want this baby?

The concept behind the Rail Gun in *Quake III Arena*, in case you're still wondering, is about firing a bullet that travels at the speed of light to zap your opponents instantly.

We can already hear cries of "Crap!" from the physics majors, but get this: a weapon like this is being worked on. DARPA has been researching



...and the University of Texas, Austin's answer

this for 10 years, and work continues. They are working on projectiles that do not contain explosives, but have extremely high kinetic speeds, which would, in theory, make their kinetic energy superior to the energy output of a shell filled with explosives of greater mass. This would then enable soldiers to carry more ammunition and also make the carrying of weapons on tank or fighter craft or boat safer. By firing their shells at far greater velocities, DARPA's "Rail Gun" will have greater range—and will not be affected by wind drift. Read: bypassing the physical limitations of conventional firearms.

Then get this: there is a working prototype at the University of Texas at Austin's Institute for Advanced Technology, capable of delivering tungsten-armour-piercing bullets with kinetic energies of nine million joules—that's like ramming a one-ton truck into a wall at 80 kmph! The only problem is that since the shells are fired at an extremely high velocity, the rail of the gun suffers from extreme damage, and has to be serviced after every shot.

Nevertheless, we can't wait to see this one in action!



Quake 4's Rail Gun...